

MODULE - 3

Measuring Instruments & Electromagnetic and Transformer

Types of Measuring Instruments

*1. PMMC (Permanent Magnet Moving Coil)

→ Used as an ammeter

→ Used as voltmeter

*2. MI (Moving Iron) Instrument

→ Attraction Type MI

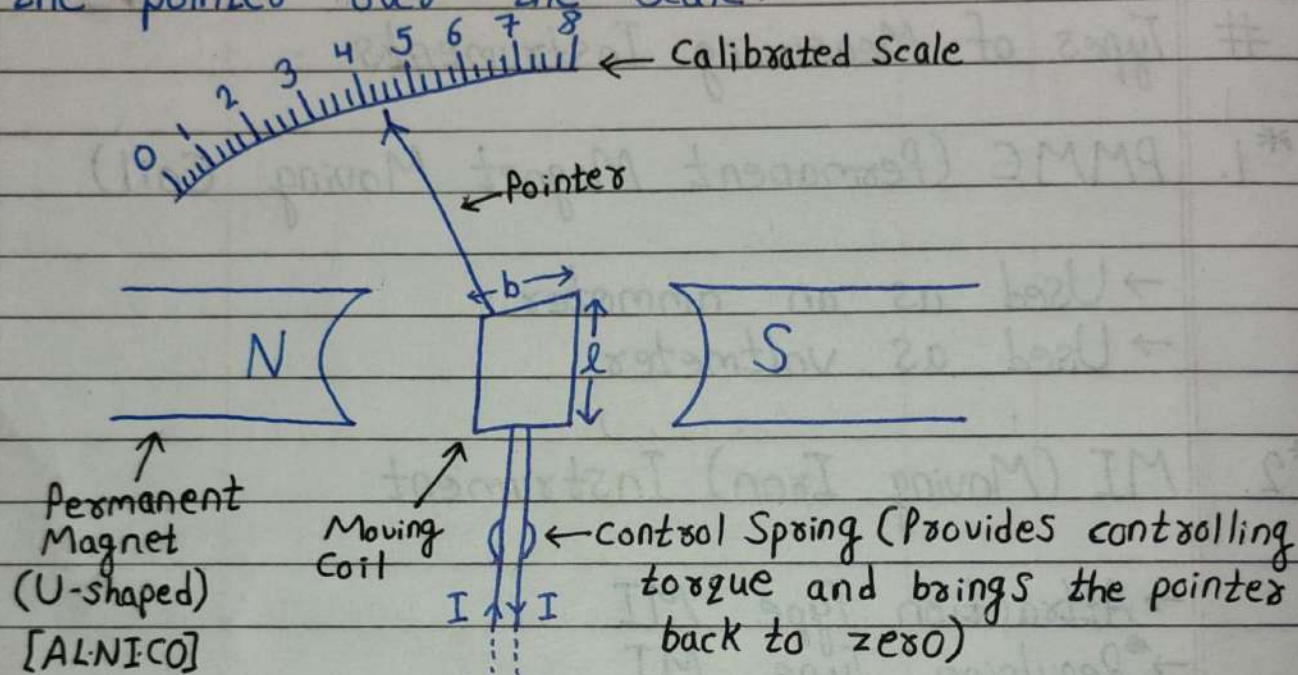
→ Repulsion Type MI

3. Dynamometer Type Instrument

*4. Single Phase Energy Meter

Imp. # PMMC (Permanent Magnet Moving Coil) or conductor

Principle: When a current carrying coil is placed in a magnetic field, it experiences a mechanical force that produces a deflecting torque. This torque moves the pointer over the scale.



When current ' I ' is passed through the coil it will experience a mechanical force ' F '.

$$F = I l B \quad [l \rightarrow \text{Length of the coil} \\ \& B \rightarrow \text{Magnetic field strength}]$$

For ' N ' no. of turns,

$$F = I l B N$$

This force produces the necessary deflecting torque.

$$T_d = F \times b \quad [b \rightarrow \text{Breadth of the coil}]$$

$$T_d = NBAI \quad [A \rightarrow \text{Area} \rightarrow l \times b]$$

If T_c is the controlling torque,

$$T_c = C\theta \quad [C \rightarrow \text{Spring Constant} \\ \theta \rightarrow \text{Deflection}]$$

At equilibrium,

$$T_d = T_c$$

$$NBAI = C\theta$$

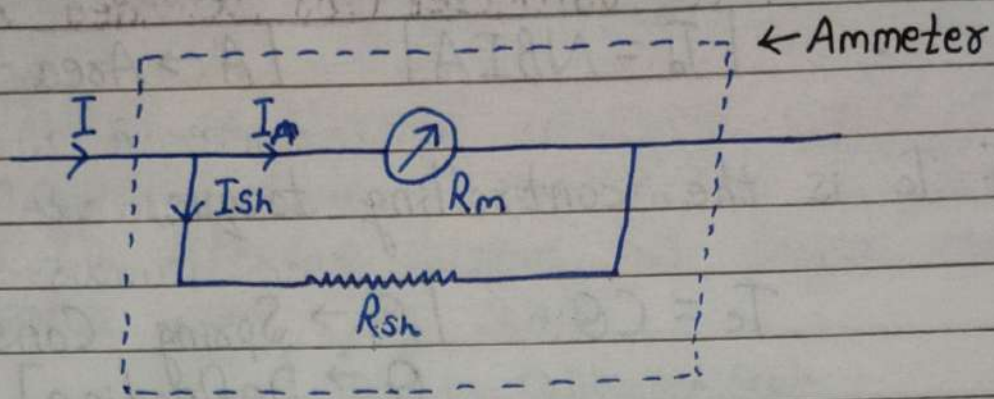
$$\theta = \frac{NBA}{C} I$$

Since, $\theta \propto I$, PMMC instrument gives a linear scale reading.

PMMC as an Ammeter (For Extended Range)

The PMMC gives full-scale deflection with a small current (microampere to milliampere).

To measure higher currents, a low resistance called a shunt (R_{sh}) is connected in parallel with the meter. This allows most of the current to pass through the shunt.



$I \rightarrow$ Total Current

$I_A \rightarrow$ Current for full-scale deflection

$I_{sh} \rightarrow$ Current through shunt

$R_m \rightarrow$ Resistance of Meter

$R_{sh} \rightarrow$ Resistance of Shunt

Resistance of ammeter,

$$R_A = \frac{R_m R_{sh}}{R_m + R_{sh}}$$

Current relation,

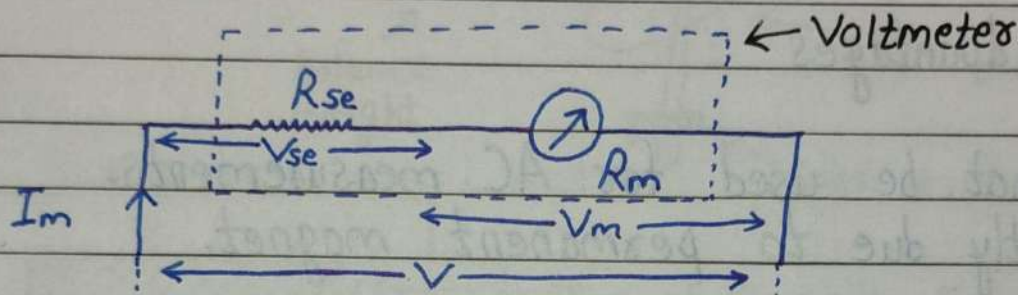
$$I = I_A + I_{sh}$$

$$I_A R_m = I_{sh} R_{sh} \quad [V_m = V_{sh}]$$

$$R_{sh} = \frac{I_A R_m}{I - I_A}$$

PMMC as a Voltmeter (For Extended Range)

To use PMMC as a voltmeter, a high resistance (R_{se}) is connected in series with the meter. This limits the current and allows measurement of higher voltages.



$V \rightarrow$ Voltage to be measured

$V_m \rightarrow$ Voltage across the meter

$V_{se} \rightarrow$ Voltage across series resistance

$I_m \rightarrow$ Current for full-scale deflection

$R_m \rightarrow$ Resistance of meter

$R_{se} \rightarrow$ Series resistance

$$V = V_{se} + V_m$$

$$V_{se} = I_m R_{se}$$

$$\Rightarrow R_{se} = \frac{V_{se}}{I_m} = \frac{V - V_m}{I_m}$$

$$\Rightarrow \boxed{R_{se} = \frac{V}{I_m} - R_m} \quad [\because V_m = I_m R_m]$$

$$R_v = R_s + R_m \quad [\text{Total Resistance of Voltmeter}]$$

Advantages of PMMC Instruments

- Uniform (Linear) scale.
- High sensitivity.
- No hysteresis loss.
- Suitable for DC measurements.

Disadvantages

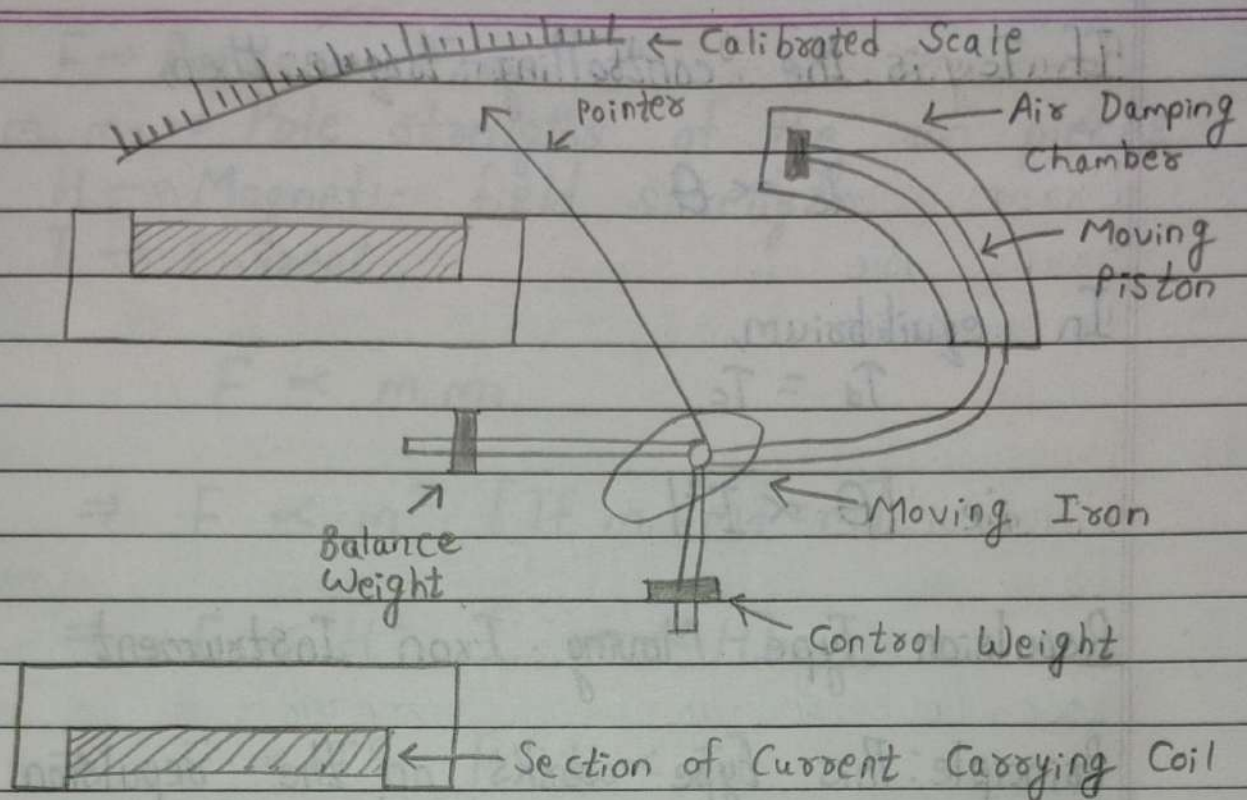
- Cannot be used for AC measurements.
- Costly due to permanent magnet.

MI (Moving Iron) Instrument

MI instruments are used to measure current and voltage in both AC and DC circuits.

Attraction Type Moving Iron Instrument

Principle: When current passes through a coil, it creates a magnetic field. A soft iron piece placed near the coil gets attracted towards the stronger magnetic field. The movement of this iron piece cause the pointer to move over the calibrated scale and show the reading.



Let, $F \rightarrow \text{Force}$

$H \rightarrow \text{Magnetic Field Strength}$

$I \rightarrow \text{Current}$

$m \rightarrow \text{Pole Strength}$

We have,

$$F \propto mH$$

$$F \propto H^2 \quad [\because m \propto H]$$

Also, $H \propto I$

$$\Rightarrow F \propto I^2 \quad \text{--- (1)}$$

We know that, deflecting torque is directly proportional to the force.

$$\text{i.e. } T_d \propto F \quad \text{--- (2)}$$

From (1) & (2),

$$T_d \propto I^2$$

If T_c is the controlling torque, then,

$$T_c \propto \theta$$

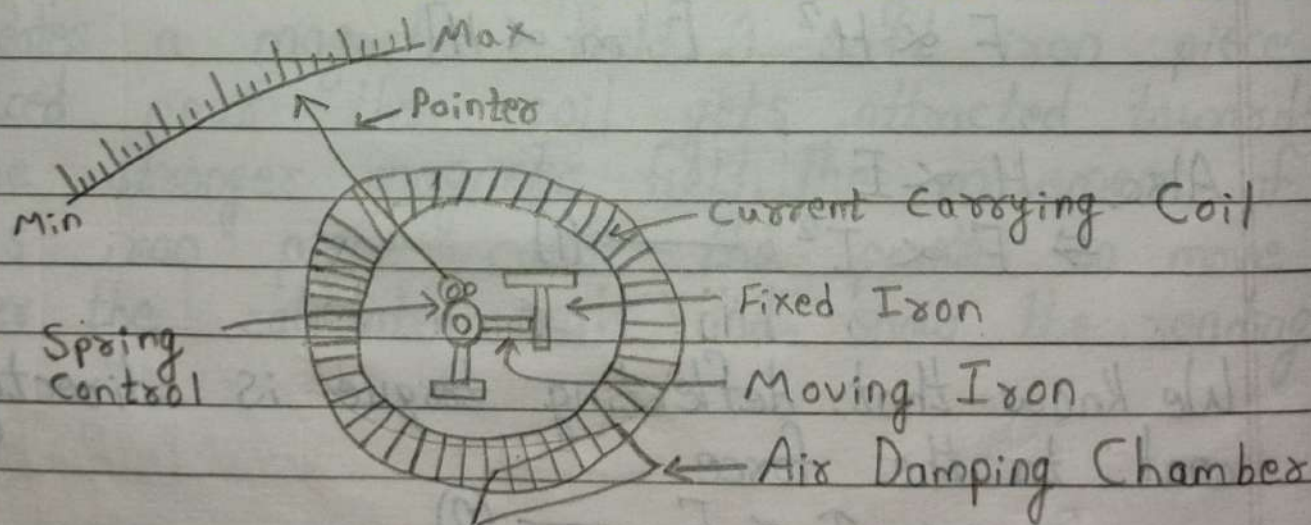
In equilibrium,
 $T_d = T_c$

i.e. $\boxed{\theta \propto I^2}$

Repulsion Type Moving Iron Instrument

Principle: This type works on the repulsion between two iron pieces magnetized with the same polarity. When current flows through the coil, it produces a magnetic field that magnetizes both the fixed and moving iron pieces.

Since, like poles repel each other, the moving iron is pushed away and the pointer attached to it moves over the calibrated scale.



Let, $F \rightarrow$ Repulsive Force

$m_1, m_2 \rightarrow$ Pole strengths of the iron pieces

$H \rightarrow$ Magnetic field strength

$I \rightarrow$ Current

$$F \propto m_1 m_2$$

$$\Rightarrow F \propto m^2 \quad [\text{If } m_1 = m_2 = m]$$

$$\Rightarrow F \propto H^2 \quad [\because m \propto H]$$

$$\Rightarrow F \propto I^2 \quad [\because H \propto I]$$

Deflecting torque,

$$T_d \propto F$$

$$\Rightarrow T_d \propto I^2$$

Controlling torque,

$$T_c \propto \theta$$

At equilibrium,

$$[\theta \propto I^2]$$

$$[T_d = T_c]$$

Advantages of MI Instruments

- Economical.
- Can be used for both AC and DC.
- Strong and simple construction.

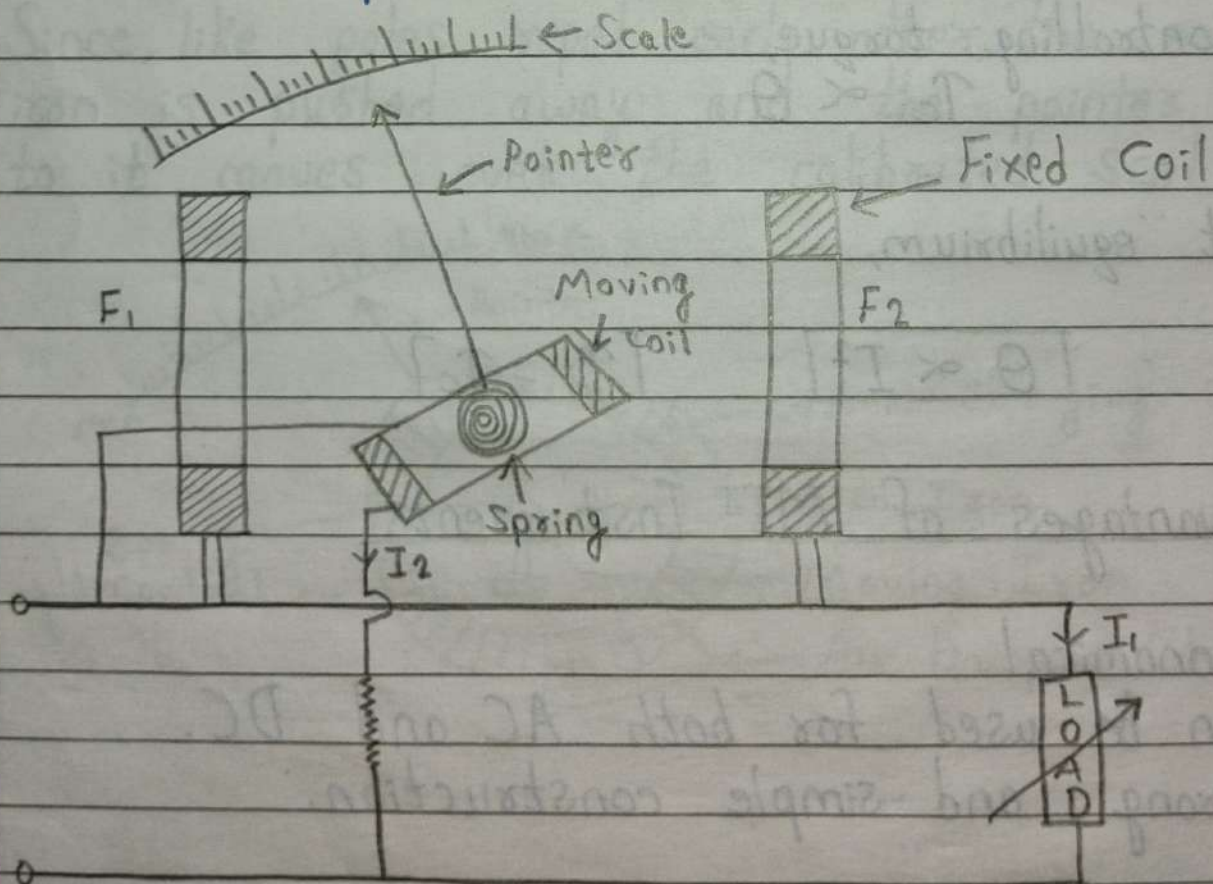
Disadvantages of MI Instruments

- Errors due to changes in frequency.
- Errors due to stray magnetic fields.
- Errors due to hysteresis effect.

Dynamometer Type Instrument

It is also called Electro-dynamic Instrument. It is commonly used as a wattmeter but can also be used as an ammeter or voltmeter.

Principle: When current flows through both the fixed and moving coils, each produces a magnetic field. The interaction between these two magnetic fields creates a deflecting torque. This torque moves the pointer over the scale.



Let, the current I_1 passed through the fixed coils F_1 & F_2 then operating field B is produced.

$$B \propto I_1$$
$$\Rightarrow B = KI_1 \quad [K \text{ is constant}]$$

Also, let current I_2 passed through moving coil then force experienced by the moving coil is given by,

$$F = BI_2 l$$

If T_d is the deflecting torque,

$$T_d = BI_2 l b$$

$$\Rightarrow T_d = BI_2 A$$

$$\Rightarrow T_d = NBI_2 A \quad [N \text{ is no. of turns}]$$

$$\Rightarrow T_d = NI_1 I_2 A \quad [\because B \propto I_1]$$

$$\Rightarrow T_d \propto I_1 I_2$$

If T_c is the controlling torque,

$$T_c \propto \theta$$

In equilibrium,

$$T_c = T_d$$

$$\Rightarrow \theta \propto I_1 I_2$$

$$\Rightarrow \theta \propto I^2 \quad [\text{If } I_1 = I_2 = I]$$

As a Wattmeter

- Fixed coil $\rightarrow$ Carries load current $\rightarrow$ Current Coil (CC)
- Moving coil $\rightarrow$ Connected across load (with high resistance) $\rightarrow$ Potential/Voltage Coil (VC).

The deflecting torque is proportional to power,

$$T_d \propto (V \times I \times \cos \phi)$$

In AC circuit,

$$T_d \propto e i$$

$$\text{or, } T_d \propto E_m \sin \omega t \cdot I_m \sin (\omega t - \phi)$$

$$\text{or, } T_d \propto \text{Instantaneous Power}$$

$$\text{or, } T_d \propto \text{Avg. Power}$$

$$\text{or, } T_d \propto \frac{1}{2\pi} \int_0^{2\pi} E_m I_m \sin \omega t \cdot \sin (\omega t - \phi) d\omega t$$

$$\Rightarrow T_d \propto EI \cos \phi$$

Advantages

- High accuracy.
- Works for both AC and DC.
- Used for power measurement (wattmeter).

Disadvantages

- Scale is not uniform.
- Errors at low power factor.
- Errors due to stray magnetic fields and eddy currents.

Imp. # Single Phase Energy Meter (Induction Type)

The energy meter is used to register or measure the electrical energy consumed in a given circuit to the given period.

There are four basic system^{or parts} regarding the construction of single phase energy meter.

1. Driving System:

It consists of two electromagnets i.e. series magnet and shunt magnet.

2. Moving System:

It consists of a movable, light, freely suspended aluminium disc which is mounted on the spindle. The moving disc is placed between the air gap of shunt and series magnet.

3. Braking System:

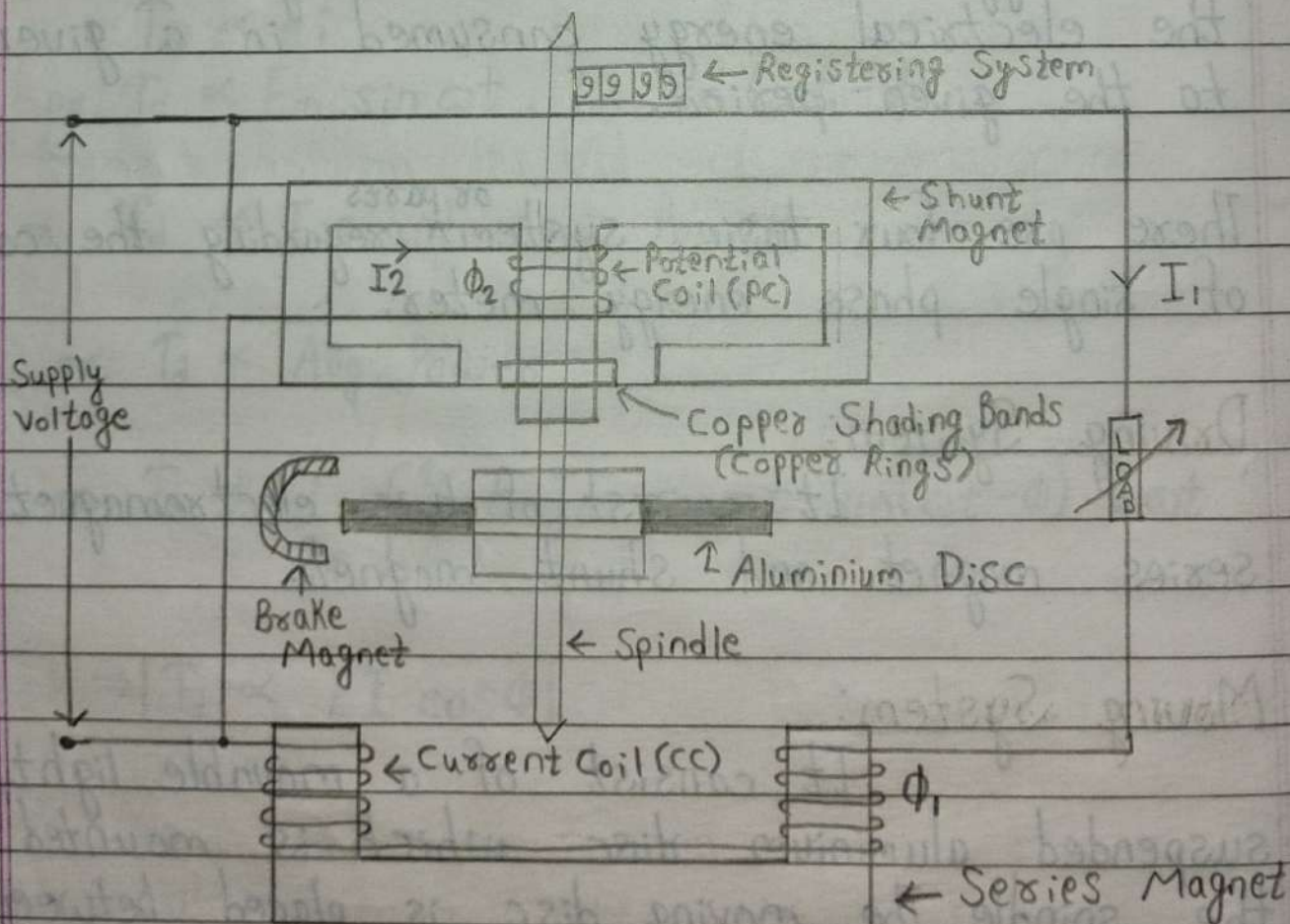
It consists of a brake magnet, which is placed near the edge of the rotating disc. It cuts the magnetic flux due to this magnet. Due to this, eddy current is induced in the disc.

4. Recording System (or) Registering System:

It consists of a mechanical counter which records the number of revolutions of the disc. The number of revolutions is proportional to the energy consumed.

The revolution of the disc is calibrated in kWh-energy consumption.

$$1 \text{ kWh} = 1 \text{ Unit} = 3.6 \times 10^6 \text{ J}$$



- The current coil produces a magnetic flux Φ_1 in phase with the line current I_1 .
- The voltage coil produces flux Φ_2 that lags the supply voltage by 90° .
- These two fluxes induced eddy currents in the aluminum disc.
- Their interaction creates a torque proportional to power.
i.e. $T_d \propto VI \cos \phi$
- The braking torque increases with speed.
i.e. $T_b \propto N$

At steady speed,

$$T_d = T_b$$

$$\Rightarrow \boxed{N \propto VI \cos \phi}$$

i.e. Number of revolutions $\propto$ Energy consumed.

Hence, the speed of the disc is proportional to power.

Advantages

- Can be used as ammeter or voltmeter.
- Low maintenance, inexpensive, simple and strong.
- Free from frequency errors.

Disadvantages

- Cannot measure DC quantities.
- Scale is non-uniform (uneven).

Application

Used for domestic and industrial energy measurement.

Q) An energy meter revolves 10 revolutions of disc for one unit of energy. Find the no. of revolution.

⇒ Number of revolutions = $10 \times (\text{Energy Consumed})$.

Transformer

Transformer is a static device that transfers electrical energy from one circuit to another circuit through the medium of magnetic induction or a magnetic field without a change in frequency.

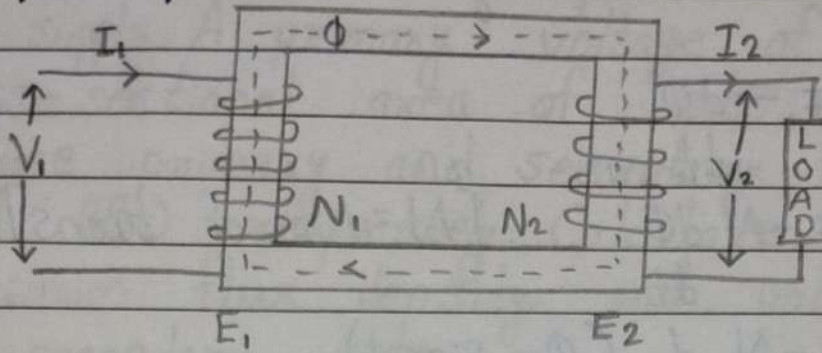
If output voltage is more than input voltage, then Step up.

$$V_2 > V_1 \Rightarrow \text{Step Up.}$$

If output voltage is less than input voltage, then Step down.

$$V_2 < V_1 \Rightarrow \text{Step Down.}$$

Principle Operation of Transformer

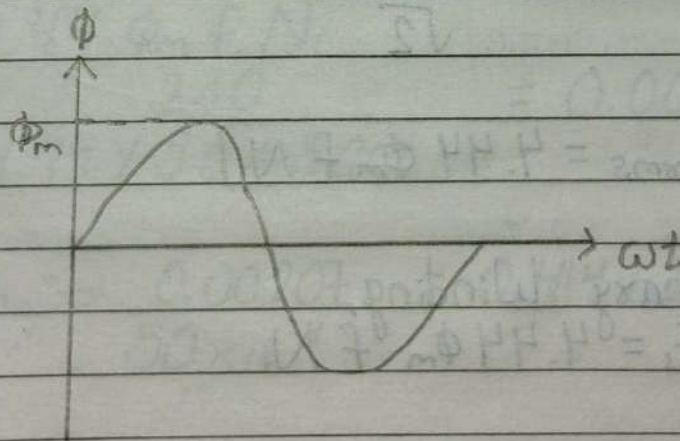


Transformer operates on the principle of mutual (electromagnetic) induction by the use of two coupled circuit or coil.

When alternating current flows through the primary coil or winding, it produces a changing magnetic field in the iron core. This varying magnetic flux induces an electromotive force (EMF) in the secondary coil (winding).

Imp.

EMF Equation for One Phase Transformer



$$\text{Let, } \Phi = \Phi_m \sin \omega t$$

According to the Faraday's law,

$$e = - \frac{d\phi}{dt}$$

$$\text{or, } e = -N \frac{d\phi}{dt} \quad [N = \text{No. of turns}]$$

$$\text{or, } e = -N \frac{d}{dt} (\phi_m \sin \omega t)$$

$$\text{or, } e = -N \phi_m \omega \cos \omega t$$

$$\text{or, } e = N \phi_m \omega \sin(\omega t - \pi/2)$$

$$\text{or, } e = e_{\max} \sin(\omega t - \pi/2).$$

where, $e_{\max} = e_m = N \phi_m \omega$

Also,

$$e_{\text{rms}} = \frac{e_m}{\sqrt{2}}$$

$$\text{or, } e_{\text{rms}} = \frac{N \phi_m \omega}{\sqrt{2}}$$

$$\text{or, } e_{\text{rms}} = \frac{N \phi_m 2\pi f}{\sqrt{2}}$$

$$\Rightarrow \boxed{e_{\text{rms}} = 4.44 \phi_m f N}$$

For Primary Winding,

$$E_1 = 4.44 \phi_m f N_1$$

For Secondary Winding,

$$E_2 = 4.44 \phi_m f N_2$$

$$\boxed{\text{Transformation Ratio} = \frac{E_1}{E_2} = \frac{N_1}{N_2} = \frac{V_1}{V_2} .}$$

Q) Step down transformer operates on a 50 Hz AC supply. A primary voltage of 230 V, the cross-sectional area of the core is 50 cm^2 . Assume primary and secondary turns are 500 and 250 respectively. Calculate maximum flux, maximum flux density and voltage induced on the secondary side.

Solⁿ:

$$f = 50 \text{ Hz}$$

$$V_1 = 230 \text{ V}$$

$$A = 50 \text{ cm}^2 = 50 \times 10^{-4} \text{ m}^2$$

$$N_1 = 500$$

$$N_2 = 250$$

$$\Phi_m = ?$$

$$B_m = ?$$

$$V_2 = ?$$

We have,

$$E_1 = 4.44 \Phi_m f N_1$$

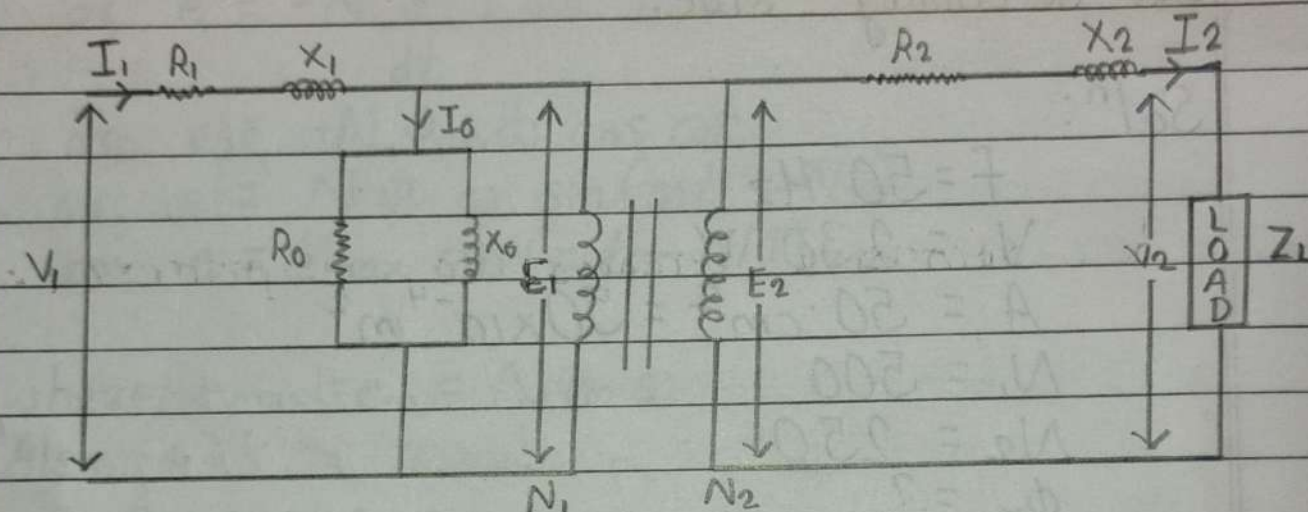
$$\therefore \Phi_m = \frac{230}{4.44 \times 50 \times 500} = 0.00207 \text{ Wb.}$$

$$B_m = \frac{\Phi_m}{A} = \frac{0.00207}{50 \times 10^{-4}} = 0.414 \text{ Wb/m}^2 \text{ or T.}$$

$$E_2 = 4.44 \Phi_m f N_2 = 114.996 \text{ V.}$$

Imp. # Equivalent Circuit of Transformer

The equivalent circuit of a transformer is a simplified model that represents its electrical characteristics using components like resistance, reactance and impedance.



Referred Quantities

When the primary winding referred to the secondary winding or vice-versa, then the components of primary side:

- ① Resistance, reactance and impedance are multiplied by square of the transformation ratio.

$$\text{i.e. } R' = k^2 R, X' = k^2 X \text{ \& } Z' = k^2 Z$$

$$\text{where, } k = \frac{E_1}{E_2} = \frac{N_1}{N_2} = \frac{V_1}{V_2}$$

- ② Voltage and induced EMF are multiplied by transformation ratio.

i.e. $E' = KE$ & $V' = KV$

- ③ Current is divided by the transformation ratio.

i.e. $I' = \frac{I}{k}$

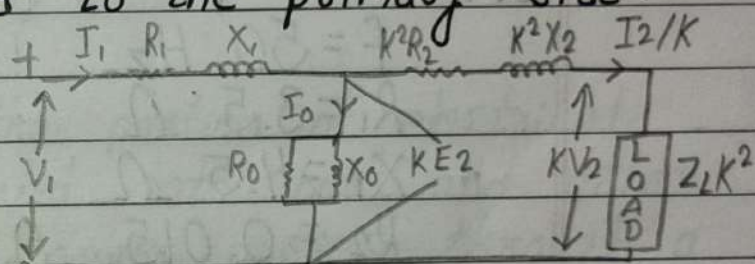
When secondary referred to the primary side:

$$R_{eq} = R_1 + k^2 R_2$$

$$X_{eq} = X_1 + k^2 X_2$$

$$Z_{eq} = R_{eq} + jX_{eq} = \sqrt{R_{eq}^2 + X_{eq}^2}$$

$$k = \frac{N_1}{N_2}$$



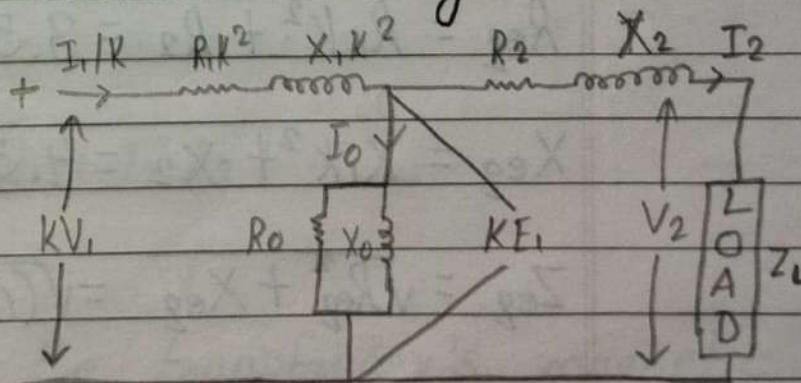
When primary referred to the secondary side:

$$R_{eq} = k^2 R_1 + R_2$$

$$X_{eq} = k^2 X_1 + X_2$$

$$Z_{eq} = \sqrt{R_{eq}^2 + X_{eq}^2}$$

$$k = \frac{N_2}{N_1}$$



9) A 2000/200 V, 50 Hz single phase transformer has primary winding resistance is 3.5Ω , reactance 4.5Ω , the secondary resistance and reactance are 0.015Ω and 0.02Ω respectively. Find equivalent resistance, reactance and impedance;

- (i) When referred to the secondary side.
- (ii) When referred to the primary side.

Solⁿ:

$$f = 50 \text{ Hz}$$

$$R_1 = 3.5 \Omega$$

$$X_1 = 4.5 \Omega$$

$$R_2 = 0.015 \Omega$$

$$X_2 = 0.02 \Omega$$

$$R_{eq} = ?$$

$$X_{eq} = ?$$

$$Z_{eq} = ?$$

$$(i) \Rightarrow k = \frac{2000}{2000} = 0.1$$

$$R_{eq} = R_1 k^2 + R_2 = 3.5 \times (0.1)^2 + 0.015 = 0.05 \Omega$$

$$X_{eq} = X_1 k^2 + X_2 = 4.5 \times (0.1)^2 + 0.02 = 0.065 \Omega$$

$$Z_{eq} = \sqrt{R_{eq}^2 + X_{eq}^2} = \sqrt{(0.05)^2 + (0.065)^2} = 0.082 \Omega$$

$$(ii) \Rightarrow K = \frac{2000}{200} = 10$$

$$R_{eq} = R_1 + k^2 R_2 = 3.5 + 10^2 \times 0.015 = 5 \Omega$$

$$X_{eq} = X_1 + k^2 X_2 = 4.5 + 10^2 \times 0.02 = 6.5 \Omega$$

$$Z_{eq} = \sqrt{R_{eq}^2 + X_{eq}^2} = \sqrt{(5)^2 + (6.5)^2} = 8.2 \Omega$$

Transformer Losses

Transformer losses are the portion of electrical energy that gets converted into heat and other non-useful forms when power passes through a transformer.

There are mainly two types of losses:

① Copper Losses (or Winding Losses):

This occurs in the coils because the copper wire has resistance. These losses increase with the load.

$$P_{cu} = I^2 R = I_1^2 R_1 + I_2^2 R_2$$

② Core (Iron) Losses:

These happen in the transformer's magnetic core. They are caused by the alternating magnetic field and include two parts.

- **Hysteresis Loss:** Energy lost as the magnetic field in the core keeps reversing direction.
- **Eddy Current Loss:** Small circulating currents induced in the core material that turn into heat.

Efficiency of Transformer

The efficiency of a transformer measures how well it converts input electrical power into output power without losses or minimum losses.

$$\eta = \frac{\text{Output Power}}{\text{Input Power}} \times 100\%$$

$$\text{Input Power} = \text{Output Power} + \text{Losses}$$

$$\eta = \frac{P_{out}(n)}{P_{out}(n) + P_{core} + P_{cu}(n^2)} \times 100\%$$

$$P_{out} = V_2 I_2 \cos \phi_2 (n)$$

$$P_{in} = V_2 I_2 \cos \phi_2 + P_i + I_2^2 R_{e2} (n^2)$$

$$\eta_{max} = \frac{(VA \text{ Rating}) \cos \phi_2}{(VA \text{ Rating}) \cos \phi_2 + 2P_i} \quad [P_i = P_{cu}]$$

$$KVA \text{ at } \eta_{max} = (KVA \text{ Rating}) \sqrt{\frac{P_i}{P_{cu}}}$$

Q) 3300 V, 50 Hz, 60 KVA, 1 ϕ transformer has 110

iron loss of 600 W, primary secondary winding resistances are 3.3 Ω and 0.011 Ω respectively. Determine efficiency of transformer at full load 0.8 pf.

Solⁿ:

$$f = 50 \text{ Hz}$$

$$P_i = 600 \text{ W}$$

$$R_1 = 3.3 \Omega$$

$$R_2 = 0.011 \Omega$$

$$\eta = \frac{60 \times 10^3 \times 0.8 (1) \times 100 \%}{(1) 60 \times 10^3 \times 0.8 + 600 + (18.18^2 \times 3.3 + 545.45^2 \times 0.011) (1)^2} = 90.62 \%$$

$$P_{cu} = I_1^2 R_1 + I_2^2 R_2$$

$$I_1 = \frac{\text{KVA (in VA)}}{V_1} = \frac{60 \times 10^3}{3300} = 18.18 \text{ A}$$

$$I_2 = \frac{60 \times 10^3}{110} = 545.45 \text{ A}$$

- Q) A 250 KVA, 1ϕ transformer has iron loss of 1.8 kW. The full load copper loss is 2000 W. Calculate
- efficiency at full load • 0.8 pf.
 - KVA supplied at maximum efficiency.
 - Maximum efficiency at 0.8 pf.

Solⁿ:

$$P_i = 1.8 \text{ kW} = 1800 \text{ W}$$

$$P_{cu} = 2000 \text{ W}$$

$$(i) \Rightarrow \eta = \frac{250 \times 10^3 \times 0.8 \times 100}{250 \times 10^3 \times 0.8 + 1800 + 2000} = 98.13 \%$$

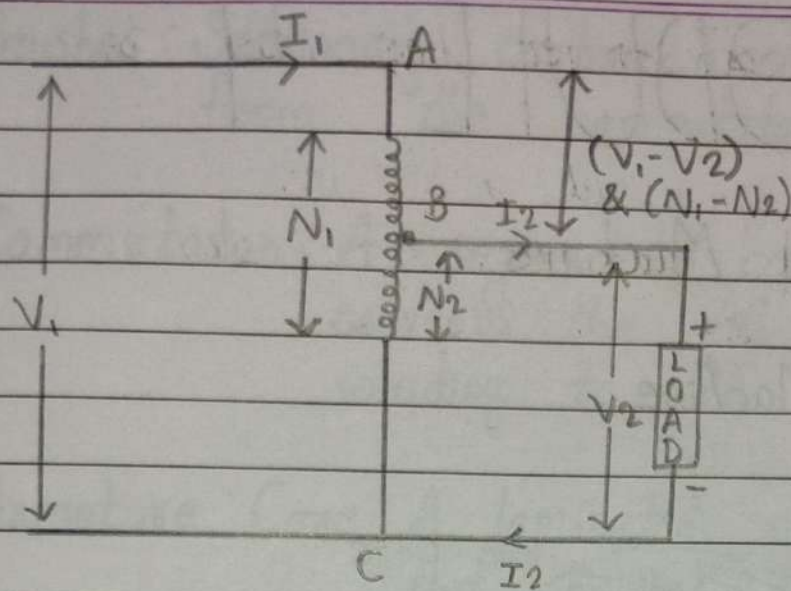
$$(ii) \Rightarrow \text{KVA at } \eta_{\max} = 250 \times \sqrt{\frac{1800}{2000}} = 237.19 \text{ KVA}$$

$$(iii) \Rightarrow \eta_{\max} = \frac{237.19 \times 10^3 \times 0.8}{237.19 \times 10^3 \times 0.8 + 2 \times 1800} \times 100 = 98.13 \%$$

Auto-Transformer

Single winding transformer is called as an auto-transformer. It works on the principle of Electro-magnetic induction.

Auto transformer is easily constructed and maintenance is easy.



In case of ideal auto transformer, when current I_1 starts flowing through the source and I_2 starts flowing at the load side, neglecting losses.

$$V_1 I_1 \cos \phi_1 = V_2 I_2 \cos \phi_2$$

$$\text{or, } V_1 I_1 = V_2 I_2$$

$$\text{or, } \frac{I_1}{I_2} = \frac{V_2}{V_1}$$

$$\Rightarrow \boxed{\frac{I_1}{I_2} = \frac{V_2}{V_1} = \frac{N_2}{N_1} = K}$$

Applications:

- Interconnection of power systems of different voltages.
- Used as variac in laboratories.
- Used as a booster transformer in transmission lines.